

NARA, Japan

WMO: 477800

Lat: 34.6737N

Lon: 135.8380E

Elev: 109

StdP: 100.02

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-1.3	-8.7	1.8	3.0	-7.0	2.1	3.9	5.4	5.0	4.9	5.2	1.2	50	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	34.2	25.1	33.2	24.8	32.1	24.5	26.1	31.9	25.6	31.1	25.1	30.3	2.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.9	28.2	24.1	19.3	27.9	23.6	18.7	27.7	81.4	32.1	79.4	31.3	77.4	30.5	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	4.2	3.5	DB	-3.8	35.9	0.7	1.1	-4.3	36.7	-4.8	37.3	-5.2	38.0	-5.7	38.8	(4)
			WB	-4.9	27.2	1.1	0.5	-5.6	27.6	-6.3	27.9	-6.9	28.2	-7.6	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	4.3	4.9	8.2	13.7	18.8	22.4	26.7	27.9	23.7	17.7	11.6	6.6
	DBStd	8.60	2.05	2.64	3.16	3.42	2.54	2.37	2.29	1.91	2.90	2.94	3.01	2.54
	HDD10.0	534	177	146	77	8	0	0	0	0	0	17	109	
	HDD18.3	1908	436	377	316	142	25	1	0	1	46	202	363	
	CDD10.0	2583	1	2	19	119	272	373	518	555	412	240	66	4
	CDD18.3	914	0	0	0	3	39	124	260	297	163	28	0	0
	CDH23.3	8057	0	0	0	42	296	791	2501	3151	1155	120	1	0
	CDH26.7	3038	0	0	0	3	48	202	1017	1392	363	13	0	0
Wind	WSAvg	1.5	1.6	1.6	1.7	1.7	1.5	1.4	1.4	1.5	1.4	1.3	1.2	1.5
Precipitation	PrecAvg	1473	53	69	108	107	150	198	177	139	195	130	89	58
	PrecMax	1856	125	142	187	225	374	368	398	375	429	299	221	113
	PrecMin	903	4	18	53	38	55	85	56	19	52	25	14	3
	PrecStd	240	30	32	37	44	79	71	92	78	88	60	52	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	17.2	21.4	26.7	30.0	32.5	35.4	36.0	33.5	28.2	22.0	16.5
		MCWB	9.4	11.7	13.0	16.4	18.6	23.6	25.6	25.3	24.6	21.6	16.2	12.5
	2%	DB	11.4	14.0	18.8	24.5	28.0	30.3	34.0	34.7	31.6	26.1	20.0	14.3
		MCWB	7.3	9.1	11.5	15.0	18.1	22.1	25.0	25.0	24.0	20.1	15.0	10.6
	5%	DB	10.0	11.9	16.5	22.4	26.4	28.9	32.9	33.7	30.3	24.6	18.5	12.9
		MCWB	6.3	7.6	10.5	14.2	17.3	21.2	24.9	25.0	23.2	18.8	14.5	9.3
	10%	DB	8.6	10.1	14.5	20.5	24.9	27.6	31.7	32.6	28.9	23.0	17.0	11.5
		MCWB	5.3	6.7	9.8	13.6	16.8	20.8	24.5	24.7	22.6	18.1	13.8	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	13.1	15.2	19.5	21.8	24.9	26.6	26.9	25.7	23.2	18.0	14.0
		MCDB	12.4	16.1	17.7	22.0	25.8	30.3	32.8	32.8	30.9	26.3	19.6	15.6
	2%	WB	8.4	10.5	13.4	17.5	20.4	23.9	26.0	26.2	25.0	21.5	16.7	11.7
		MCDB	10.1	12.5	16.6	21.1	23.9	27.8	31.9	32.1	29.7	24.7	18.4	13.4
	5%	WB	6.9	8.6	11.9	16.0	19.3	23.1	25.5	25.7	24.3	20.2	15.5	10.0
		MCDB	9.1	10.7	15.0	20.2	23.3	26.7	31.0	31.4	28.3	23.0	17.7	12.0
	10%	WB	5.7	7.0	10.4	14.7	18.3	22.3	25.0	25.3	23.6	18.9	14.2	8.6
		MCDB	8.2	9.7	13.8	18.8	22.6	25.7	30.2	30.7	27.2	21.7	16.5	10.9
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	10.4	11.3	10.8	8.7	8.3	9.1	8.6	9.1	9.0	8.2
		MCDBR	10.1	12.3	13.8	14.4	13.7	11.2	10.2	10.6	9.9	10.5	10.4	10.1
		MCWBR	6.7	7.7	7.9	6.8	5.8	4.4	3.5	3.4	4.0	5.2	6.3	6.7
	5% WB	MCDBR	9.1	10.1	11.2	10.5	9.7	8.2	9.2	9.2	8.5	8.4	8.5	8.7
		MCWBR	6.8	7.5	7.8	6.5	5.5	3.9	3.6	3.4	3.7	4.9	6.0	6.7
Clear-Sky Solar Irradiance	taub	0.402	0.442	0.502	0.536	0.541	0.560	0.562	0.529	0.486	0.448	0.417	0.397	
	taud	2.146	2.049	1.909	1.877	1.931	1.955	2.002	2.083	2.171	2.194	2.189	2.173	
	Ebn at Noon	773	783	768	765	767	750	745	763	777	771	752	751	
	Edh at Noon	120	147	183	198	190	185	176	160	140	126	114	110	
All-Sky Solar Radiation	RadAvg	2.30	2.95	3.93	4.76	5.18	4.66	4.96	5.09	4.00	3.23	2.50	2.09	
	RadStd	0.16	0.23	0.29	0.44	0.55	0.44	0.65	0.45	0.35	0.33	0.21	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page